

CTSpinoPelvic1K: spine, pelvis, ribs and femora in one coordinate frame, annotated for lumbosacral transitional anatomy

Article type: Dataset and Software Article

Authors

1. Gregory Schwing, MD, PhD ¹
2. Ashley Schehr, BS ²
3. Annika Tekumulla, BS ²
4. Margret Khoushi, BS ²
5. Ryan Christian, BS ²
6. Dane Hubers, BS ²
7. Faris Mahjoub, BS ²
8. Hassan Saad, BS ²
9. Mia Sooch, BA ²
10. Sathya Siddapureddy, BS ²
11. Michael McLellan, BS ²
12. Jerick Kim, BS ²
13. Miraziz Ismoilov, MD ³
14. Nizar Alnabahneh, MD ³

¹Department of Surgery, Detroit Medical Center and Wayne State University, Detroit, Michigan, United States of America

²School of Medicine, Wayne State University, Detroit, Michigan, United States of America

³Department of Radiology, Detroit Medical Center and Wayne State University, Detroit, Michigan, United States of America

Corresponding author

Gregory Schwing

Department of Surgery, Detroit Medical Center and Wayne State University

Detroit, Michigan, United States of America

Email: gregory.schwing@med.wayne.edu

Author contribution statement

G.S. conceived the dataset and was its intellectual lead: he designed the label scheme, implemented the processing pipeline and the quality-control procedures, and wrote the manuscript. A.S. led the data annotation, annotated data, and edited the manuscript. A.T., M.K., R.C., D.H., F.M., H.S., M.S., S.S., M.M. and J.K. annotated data. M.I. and N.A. performed the Castellvi grading, each reading every case independently and resolving disagreements by consensus, adjudicated the candidate instrumentation, and trained the student annotators in CT reading. All authors read and approved the final version of the manuscript.

Conflict of interest statement

The authors have no relevant conflicts of interest to disclose.

Acknowledgements

The imaging analyzed here was made available by The Cancer Imaging Archive and by the investigators who published the source annotations. This work received no funding.

Data and code availability (unblinded)

Dataset, archived before submission: <https://doi.org/10.5281/zenodo.22139642> (concept identifier, resolves to the current version); the version described in the manuscript is v8, <https://doi.org/10.5281/zenodo.22293878>.

Code: <https://github.com/Gregory-Schwing-MD-PhD/CTSpinoPelvic1K>, tagged at the release commit.

Mirror: <https://huggingface.co/datasets/OpenSpineConsortium/CTSpinoPelvic1K> — a convenience mirror, not the archive of record.

Ethics

This work uses publicly available, de-identified imaging and annotations derived from it. The Wayne State University Institutional Review Board issued a Notice of IRB Administrative Determination of Non-Human Participant Research on 2 September 2026 (WSU IRB HPR Number 2026-269, “Open Spine Consortium: Secondary Analysis of Publicly Available, De-Identified Imaging Datasets and the Derived Annotations, Software, and Models Produced From Them”), finding that the project does not constitute human participant research under the Common Rule at 45 CFR 46 and that IRB review and oversight are not required.

**CTSpinoPelvic1K: spine, pelvis, ribs and femora in one coordinate frame, annotated
for lumbosacral transitional anatomy**

[Author names removed for double-anonymized review]

[Affiliations removed for double-anonymized review]

(Dated: 6 September 2026)

Purpose: For transitional anatomy the gold standard for numeration remains whole-spine imaging and counting caudally from C2¹. Abdominal CT does not contain C2, and where a lumbosacral transitional vertebra (LSTV) alters the count the local anatomy is ambiguous: four rib-free lumbar vertebrae may mean an L1 bearing a lumbar rib or an L5 assimilated to the sacrum, and six may mean a sixth lumbar vertebra, a T12 whose ribs are aplastic, or an S1 lumbarized from the sacrum². CT-SpinoPelvic1K is built so that this can be asked of the data: whether local morphology resolves them without counting from C2. It provides 802 CT records anchoring each lumbar level on the lowest rib-bearing vertebra above and the first sacral segment below, with radiologist-sourced spine *and* pelvic annotation on one frame, per-level ribs and femora. Classes for a sixth lumbar vertebra, a thirteenth thoracic vertebra, a separate first sacral segment and lumbar ribs cover the anomalies.

Acquisition and Validation Methods: Records pair TCIA CT COLONOGRAPHY imaging with CTSpine1K³ vertebral and CTPelvic1K⁴ pelvic labels on the acquisition with highest bone coverage, under a VerSe-native scheme. Validation: geometric invariants (802/802 pass); rib-vertebra incidence across 5,749 evaluable ribs (2 offsets, 0.035%); and agreement of spinopelvic measures with published values.

Data Format and Usage Notes: Gzipped NIfTI image/label pairs sharing one affine, canonicalized to PIR, with frozen patient-grouped LSTV-stratified five-fold splits and a reference loader; archived under a persistent identifier, <https://doi.org/10.5281/zenodo.22139642>.

Potential Applications: Testing whether a network can classify a vertebra from local features alone; updating textbook morphometry drawn from small cadaveric series; spinopelvic assessment; variant studies; opportunistic screening. *Limitations:* thoracic ground truth is field-of-view limited; postural angles supine; no held-out test set; the rib layer is a triaged-review pseudolabel; Castellvi grades are a two-reader consensus.

6 I. INTRODUCTION

7 The spine commonly carries seven cervical, twelve thoracic and five lumbar vertebrae, and
8 a vertebra is identified by where it falls in that sequence. Transitional variants sit at the
9 two ends of the lumbar column, the thoracolumbar transitional vertebra (TLTV) above and
10 the lumbosacral transitional vertebra (LSTV) below, and they disturb identification in two
11 ways at once. They change what the border vertebra and its ribs *look like*: a thoracolumbar
12 border vertebra may carry a full rib, a hypoplastic stump, or none at all, and a lumbosacral
13 one may be partly or completely assimilated to the sacrum (*sacralization*), with an anteriorly
14 wedged body, a reduced disc beneath it and hypoplastic facet joints, or conversely separated
15 from it (*lumbarization*), with a squared body, lumbar-type facets and a full-height disc².
16 They also change *how many* vertebrae each region contains: eleven or thirteen thoracic,
17 four or six lumbar.

18 Either change alone would be tractable. Together they make it difficult to state defini-
19 tively what any vertebra at these borders *is*. Four rib-free vertebrae above the sacrum may
20 mean an L1 bearing a lumbar rib or an L5 assimilated to it; six may mean a true sixth
21 lumbar vertebra, a T12 whose ribs are aplastic, or an S1 lumbarized away from the sacrum.
22 Each set yields the same count, and the morphology that would separate the readings is not
23 decisive to a human reader in every case. Whether it differs enough for a learned classifier
24 to separate them, or at least to assign each reading a probability, is the question this release
25 is built to support. Absent that, what remains is the count itself: the gold standard for
26 numeration in transitional anatomy is whole-spine imaging, counting caudally from C2¹, so
27 that each vertebra is named by its ordinal position whatever it carries, and no spine-limited
28 acquisition can supply it. LSTV is common, reported between 4% and 30% depending on
29 definition, and wrong-level surgery is its most serious consequence.

30 Existing public collections cannot address this, and size is not the reason (Table I). Spine
31 datasets omit the pelvis; pelvic datasets do not number the vertebrae; and TotalSegmentator,
32 the whole-body scheme, has no class for a sixth lumbar vertebra or for a rib on a lumbar
33 vertebra. A six-lumbar spine therefore cannot be recorded in any of them, however good the
34 segmentation, and a segmenter trained on them inherits the gap: faced with an enumeration
35 anomaly it must shift the whole column by a level or absorb a vertebra into its neighbor.
36 The size of that problem has now been measured on CT, where prior labeling methods

37 assigned every vertebra correctly in 77% of subjects and an anomaly-aware method raised
38 that to 99%⁵. VerSe^{6,7} makes the point most clearly (Table I): it deliberately enriched for
39 anatomical variants and graded them by Castellvi, yet left the graded vertebrae fused to the
40 sacrum unsegmented.

41 The intraoperative side of the problem has prior art of its own. LevelCheck registers
42 the intraoperative radiograph to the preoperative CT and projects the CT’s vertebral labels
43 onto it, localizing the target level in a limited field of view to within a few percent of cases
44 in clinical use^{8–10}. The labels it projects are placed by hand on the CT, by a skilled operator
45 and verified by the surgeon⁸: the method assumes a correctly labeled CT and does not
46 supply one. Automating that step, and getting it right precisely where the anatomy is
47 anomalous, is what a dataset carrying the classes those anomalies produce is for.

48 CTSpinoPelvic1K places the spine, pelvis, ribs and femora in one coordinate frame and
49 gives a class to each structure an enumeration anomaly produces: a sixth lumbar verte-
50 bra, and a rib borne by a lumbar vertebra. A scheme without those classes lacks a class
51 vocabulary expressive enough to describe anatomic anomalies.

52 *Where the material is.* Everything described here is v8 of a single archived dataset,
53 deposited at <https://doi.org/10.5281/zenodo.22293878>. Citations should use <https://doi.org/10.5281/zenodo.22139642>, the permanent identifier for the dataset across all
54 of its versions, which resolves to whichever version is current; the version identifier above
55 pins the exact release measured in this manuscript. Documentation, the label scheme, the
56 reference loader and the per-record quality-control tables are distributed with the archive
57 and are described in Sec. III.

59 I.A. Vertebra labeling when the scan cannot settle the count

60 *The limitation, stated plainly.* No scan in this corpus contains C2, so the conventional
61 count cannot be performed: a property of abdominal imaging, not of the annotation. A
62 thirteenth thoracic vertebra, a lumbar rib and a *stump rib* (a hypoplastic twelfth rib, a
63 cardinal indicator of a thoracolumbar transitional vertebra) leave the same number of rib-
64 free vertebrae between the two borders of the lumbar spine, the last rib-bearing vertebra
65 and the sacrum.

66 *What the release does in addition.* Every vertebra carries the identifier its radiologist-

TABLE I Public CT collections at the lumbosacral junction. Black, class present; outlined, graded only to exclude it⁷.

Collection	Scans	Count	L6	Sacrum	S1	Pelvis	Ribs	L. rib	Femora	Grade
CTSpine1K ³	1,005									
CTPelvic1K ⁴	1,184									
VerSe ⁶	374									
RibSeg v2 ¹⁶	660									
TotalSegmentator ²⁰	1,204									
CTSpinoPelvic1K	802									

sourced annotation assigned, corrected where the source was wrong; those labels are the ground truth. Alongside them, every released measurement is also expressed against the two landmarks present in essentially every abdominal scan, the sacrum below and the lowest rib-bearing vertebra above, so that the measurement is positional rather than nominal: a vertebra by how many rib-free vertebrae separate it from the sacrum, a lowest rib by its length as a fraction of the rib above it, a transverse process by its span and its distance from the ala. None of these changes according to whether a reader would have said L5 or S1, so cases that disagree about the name remain comparable, and where the junction is in view the count is determined and the two descriptions agree.

But the vertebrae themselves are known to differ. Counting is not the only route to an identity. Level-specific morphometry is long established: pedicle width and height change systematically from the thoracic to the lumbar spine,¹¹ vertebral body, endplate and canal dimensions differ by level,^{12–14} and the transverse process and the iliolumbar ligament mark the last lumbar vertebra independently of any count.^{2,15} If those differences hold at the boundary itself, the phenotype is decidable from the vertebra in front of the reader without any global count. On this corpus it does hold: separating T12 from L1 on shape alone, with each patient’s own size divided out and cross-validation grouped by case, gives an area under the curve of 0.990 and 97.3% accuracy over 1485 vertebrae. Size is removed because raw millimeters separate thoracic from lumbar trivially and teach nothing about the junction, where the two are nearly the same size.

That is reported as a property of the released measurements rather than as a method:

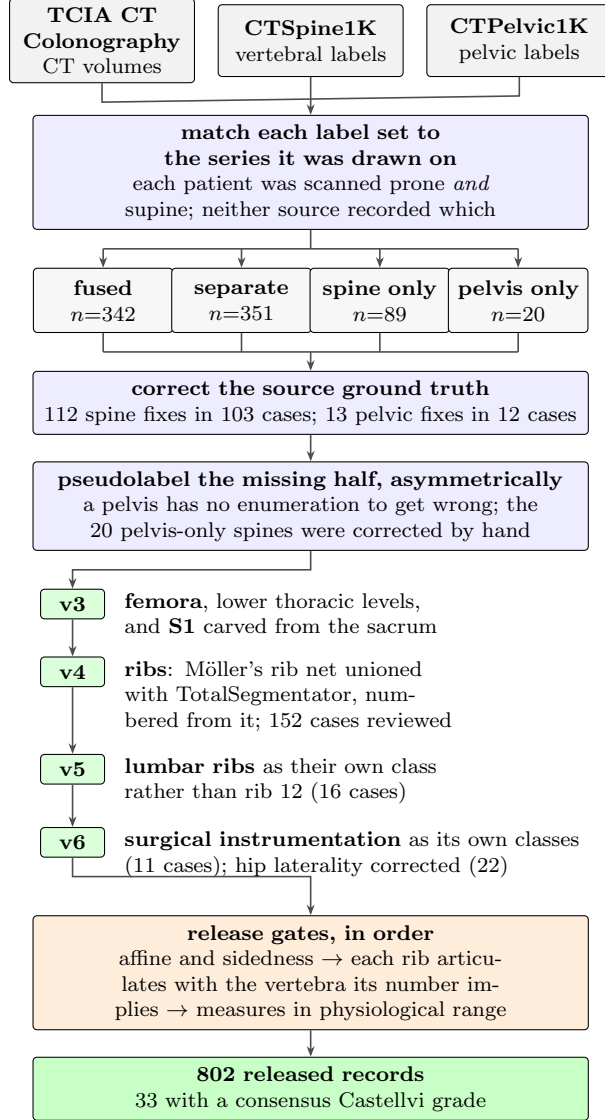


FIG. 1 How the dataset was built. n counts scans, or corrections, at each step.

the information needed to settle these variants is present locally.

II. ACQUISITION AND VALIDATION METHODS

II.A. Sources, and why a crosswalk was necessary

Both CTSpine1K³ and CTPelvic1K⁴ draw part of their imaging from the TCIA CT colonography collection^{17,18}, and the vertebral and pelvic annotations in both were produced under board-certified radiologist supervision. Neither released the mapping from an annotation to the specific CT series it was drawn on.

95 That mapping matters because every patient in CT COLONOGRAPHY was scanned
96 *twice*, prone and supine, and an annotation carries a patient identifier and nothing finer.
97 Turning a patient from supine to prone alters lumbar lordosis and the alignment of each
98 segment against the next, so outlines drawn on one series are the wrong *shape* for the other
99 and no rigid realignment recovers them. Such a mask looks competent in isolation and
100 reveals itself only overlaid on the image. The crosswalk (Fig. 1) therefore resolves each
101 annotation to a patient and then, within that patient’s volumes, to the series whose bone
102 *agrees* with it: candidates are thresholded at bone attenuation and scored by overlap with
103 the mask.

104 II.B. Fused and split records

105 CTSpine1K and CTPelvic1K each annotated a patient independently, so their labels are
106 not guaranteed to come from the same CT series (Fig. 1). This turned out to matter for
107 351 patients, nearly half the cohort: the two collections had labeled different acquisitions of
108 the same person, one prone and one supine, with neither recording which. Because posture
109 changes spinal alignment, a spine label from one acquisition cannot simply be paired with a
110 pelvic label from the other. These 351 records are released but held out of any analysis that
111 assumes a single posture. Two labeled acquisitions of the same patient in different postures
112 are, instead, a resource for studying postural alignment, not a defect.

113 II.C. Densification, and why it is asymmetric

114 A corpus in which half the records lack a pelvis is not a spinopelvic dataset, so the missing
115 structures were pseudolabeled, asymmetrically (Fig. 1). Pelvis onto spine-only records is the
116 safe direction: the sacrum and innominate bones carry no enumeration to get wrong, and
117 quality is reported as held-out performance on the *pelvic_only* records, withheld from training
118 for that purpose. The reverse direction carries the whole problem this dataset addresses,
119 since a pseudolabeled spine must commit to a count and on a transitional vertebra commits
120 to the commoner reading; those 20 records were corrected by hand, tractable at that scale.

II.D. Ribs

No public rib annotation covers this cohort, and the two available automatic sources fail in complementary ways.

Möller’s binary rib network¹⁹ reports high accuracy, and its authors show that stump ribs can be classified from a partial field of view; that result concerns the morphological features computed on a rib once segmented. On this corpus the segmentation itself failed on ribs that were *partially outside* the field of view: a rib crossing the boundary of an abdominal acquisition was liable to be missed altogether rather than segmented to the edge. This is not a limitation reported by its authors: their evaluation excluded subjects in which the last rib was not visible, and their failure analysis attributes errors to costal-process fusion, so it is recorded here as an observation from this cohort. Since this is a colonography cohort, the lower ribs are routinely clipped, and those are precisely the ribs the count depends on.

TotalSegmentator²⁰ does segment clipped ribs and supplies the per-rib numbering that a binary network cannot. Its characteristic failure is at the other end of the bone: the most medial portion of the rib (roughly the last tenth to sixth approaching the costovertebral joint) is frequently absent. A rib truncated at its medial end never reaches the vertebra it articulates with, so the rib–vertebra incidence check (the quality-control gate on the rib class labels, which matches each rib to the vertebra its number implies) has nothing to test.

The two were therefore unioned (Fig. 1): TotalSegmentator supplies numbering and the medial-clipped cases, Möller supplies detection where its output is more complete, and the union carries TotalSegmentator’s per-level identities.

The result is a pseudolabel whose review was triaged rather than exhaustive, which is worth stating plainly because a layer described only as “human-corrected” invites the reader to assume every bone was looked at. A label-based rule flagged every record in which one rib bone carried more than one identifier, or in which a rib failed to reach the vertebra it articulates with, since a single bone carrying two numbers is a numbering error by construction, whatever the anatomy. The rule selected 152 records; 148 were reviewed and corrected, and 4 were not and are identified in the release. Reviewers were medical students working through a student research consortium²¹, trained against a written labeling protocol, with every correction recorded against the annotator who made it. Records passing the rule were not individually inspected, so the layer only guarantees freedom from the failure modes

the rule detects, no more. The residual rib-vertebra offsets reported in Sec. II.G are the independent check on what the rule missed.

II.E. Label scheme and counting anchors

The scheme is VerSe-native: vertebrae keep their VerSe identifiers (C1–C7 = 1–7, T1–T12 = 8–19, L1–L6 = 20–25, sacrum 26), and every non-VerSe structure takes a fixed identifier above that range: S1 carved from the sacrum (29), hips (30–31), femora (32–33), ribs (34–45 left, 46–57 right), lumbar ribs (74–75) and surgical hardware (76–82). Identifiers 58–73 are unassigned: a soft-tissue block once reserved there was never populated and is retired.

Two anchors make the count decidable, and Fig. 2 shows both: the lowest rib-bearing vertebra above, **S1** carved from the sacrum below. Each earns its place by what it supports rather than by being an endpoint. S1 supplies the sacral endplate landmark that sacral slope and pelvic incidence are measured from, and with both brackets present L5 and L6 are distinguishable where a spine-limited view alone would not be.

How S1 is cut. The sacrum’s outer boundary is the source annotation and is never altered; S1 is the cranial part of that same bone, separated by a plane. The plane’s normal is the sacrum’s own cranio-caudal axis, made orthogonal to a left-right axis measured directly rather than assumed, so as to remove the patient’s rotation within the scanner. That rotation is not negligible: median 4.5°, maximum 16.1°, and 508 of 801 records above 3°. The level of the cut preserves the sub-division volume of the preceding release, so the plane changes the shape and orientation of the S1/S2 boundary and not how much of the sacrum is called S1. Per-record geometry ships with the archive.

Neither anchor is novel on its own: TotalSegmentator²⁰ carries a separate S1 class and per-level ribs, and VerSe⁶ carries L6, whose identifier this release keeps, so its L6 *is* VerSe’s. L6 is retained because a scheme without it cannot record a six-lumbar spine at all.

A rib on a lumbar body receives its own class, and this is the one class here with no published counterpart. A lumbar rib and a stump rib are the same object under two counts, so assigning it a number would commit the label to one reading of an enumeration anomaly. A scheme that numbers every rib 1–12 has nowhere to put a thirteenth: the annotator must either call it rib 12, asserting the vertebra beneath it is thoracic, the very question at issue, or discard it. TotalSegmentator carries ribs 1–12 per side and no such class, and VerSe’s

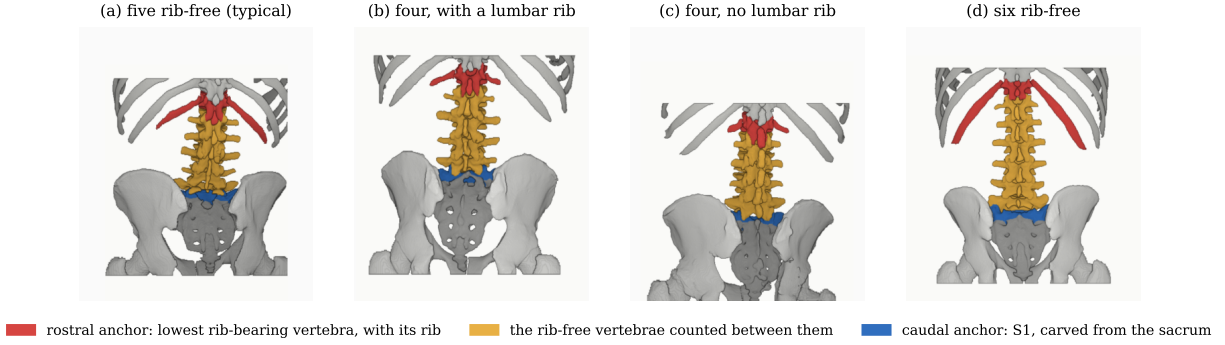


FIG. 2 The two anchors, from the released labels. Red, lowest rib-bearing vertebra and its rib; blue, S1; yellow, the rib-free vertebrae between. (b) and (c) both count four.

182 T13 is a vertebra rather than a rib. This scheme can record both: VerSe’s T13 (28) for a
 183 thoracic-type thirteenth vertebra with a true rib, and 74/75 for a lumbar-type body bearing
 184 a rudimentary one, so either reading is recorded as itself. No record here carries a T13; the
 185 sixteen lumbar-rib cases were read as lumbar-type bodies. Where a border vertebra could
 186 not be settled by its readers the label stands as read, and reclassifying those records with
 187 a classifier trained on local vertebral morphology (Sec. V) is the intended path for a later
 188 release.

189 Figure 2 shows why the interval is the primitive and the label is derived. Among the
 190 records with four rib-free vertebrae, some reach that count through a lumbar rib and some
 191 without one; the two are indistinguishable by the count alone and are distinguished here
 192 only because rib presence is itself labeled.

193 Sixteen records carry a lumbar rib: thirteen bilateral and three unilateral, all on the
 194 right, a count too small to support any statement about side preference.

195 Eighteen records carry an L6. Fourteen are labeled lumbarization, two sacralization,
 196 one semi-sacralization and one normal, which is worth stating because it shows the two
 197 annotation axes disagreeing in the open: a record can carry six lumbar-type vertebrae and
 198 still be read as a sacralization, or as unremarkable, depending on where the reader started
 199 the count.

200 Both counts are derived from the released label volumes, and so are the manifest fields
 201 that report them. That is a correction: through v5, `has_l6` was true for one record that
 202 contains no L6 and false for all eighteen that do, `has_lumbar_rib` was false everywhere, and
 203 `n_lumbar_labels` read zero in 799 of 802 records. The fields are rebuilt for this release by

counting identifiers 25 and 74–75 in each volume, with `lumbar_rib_side` added alongside them.

II.F. Computational tools

Access. Every processing step described above is performed by code released under an open-source licence and archived with the dataset; no step depends on software that is not publicly obtainable. The release archive carries the reference loader, the label-scheme definition and the quality-control scripts that generated the tables in this manuscript.

Versions and parameters. Segmentation of femora, the sacral sub-division and the per-level rib numbering used TotalSegmentator²⁰ (`total` task, release 2.x) on a single graphics processing unit. The binary rib network is Möller’s¹⁹ released weights, applied through the nnU-Net v2 framework²². The pelvic completion for records whose pelvis was absent used a five-fold nnU-Net v2 ensemble, applied out-of-fold: each record is completed by the fold that never trained on it, so no record is completed by a model that has seen it. The five fold checkpoints, with their nnU-Net plans and dataset descriptor, are released alongside the data (<https://huggingface.co/anonymous-mlhc/spinopelvic-seg-checkpoints>). Image handling throughout used NiBabel and SciPy; no image is resampled or reoriented when a label is written, so every label shares its image’s grid and affine exactly.

Generative artificial intelligence. Large language models assisted with drafting and editing text and with analysis and figure code, under author review. They were not used to generate, annotate or interpret imaging data; every number reported here is computed by the released code from the released volumes.

II.G. Validation

Validation is layered, and the ordering matters: measurements computed on a corpus that has not established internal consistency do not look broken, they look finished.

Geometric invariants. Every case is checked for label geometry agreeing with its CT in shape, affine and voxel spacing; for identifiers within the published scheme; for a non-empty label; and for sided structures falling on the correct sides, with the left–right axis read from the affine rather than assumed. The check exists because a renumbering pass once wrote

232 a reoriented array under a canonical affine and transposed a label away from its image, a
233 failure invisible downstream.

234 The check compares *each sided pair separately*: testing the ribs alone passed all 802
235 records while four carried `left_hip` on the patient’s right, and pooling all sided structures
236 into one test still missed one of those four, because a large correctly-sided structure masks
237 a smaller transposed one. A gate that passes everything is not evidence of a clean corpus
238 until it has been shown capable of failing.

239 *Rib–vertebra incidence*. Each rib is matched to its nearest vertebra and compared against
240 the vertebra its number implies. Of 11,548 ribs, 5,788 are not evaluable because the expected
241 vertebra lies outside the field of view and 11 have no vertebra within the anchor distance;
242 of the 5,749 evaluable, 5,747 match, two (0.035%) are offset by one level, and no case is
243 misnumbered. The denominator is stated because quoting the two offsets against all 11,548
244 ribs would halve the rate by counting ribs the check never examined. Both offsets are
245 field-of-view truncations on individually characterized cases, one at +1 level and one at −1,
246 reported rather than suppressed.

247 The same check carries an invariant that reads as a null result. Of the 5,749 evaluable
248 ribs, *zero* are assigned to a lumbar vertebra, not because no lumbar ribs exist, since 16
249 records carry one, but because such a rib takes class 74 or 75 and never enters the numbered
250 set the test examines. A numbered rib landing on a lumbar vertebra would be a counting
251 error of exactly the kind this dataset exists to expose.

252 *Agreement with published values*. Derived spinopelvic measures are compared against
253 Vialle *et al.*²³ (Table II). Pelvic incidence is the strongest of these checks because it is a
254 morphological property of the pelvis rather than a posture, and so is directly comparable to
255 a standing reference cohort.

256 II.H. Castellvi typing

257 Every record whose vertebral count departs from five carries a Castellvi grade²⁴ in the
258 released manifest: 33 cases, typed I–IV with the *a/b* unilateral–bilateral qualifier. This is
259 released as an annotation layer because the grade and the count are *different axes* and the
260 dataset is built to show that they are: grade IIIb occurs here at rib-free counts of four, five
261 and six alike, and seven of the 33 graded cases carry a perfectly normal count of five. A

TABLE II Derived measures against a standing reference ($n = 260$)²³; this cohort is supine.

Measure	This dataset (supine)	Reference (standing)
Pelvic incidence	54.7°	54.7 ± 10.6
Pelvic tilt	17.3°	13 ± 6
Sacral slope	36.9°	41.0 ± 8.4
Lumbar lordosis (supine)	52.7°	43 ± 11.2 (standing)
PI – LL mismatch	2.6°	centered near 0

corpus that recorded only the count would describe those seven as unremarkable.

Two radiology resident physicians graded every case independently and resolved their six disagreements by consensus; both reads and the consensus are in the manifest. The distinction the classification turns on, bony fusion against articulation without it², is the one carrying clinical weight, and it is where the disagreements lay: four of the six were III against II.

II.I. Surgical hardware, and why a threshold is not enough

Metal is trivial to detect and hard to interpret, and the difference matters here because **an iatrogenic fusion is indistinguishable from a congenital one to a distance measurement**: a cage-bridged interspace reads as “no gap” exactly as a fused transitional vertebra does. An unrecorded instrumented case is therefore a false positive on the central measure.

II.I.0.a. Detection over-calls by a factor of five. A 2500 HU threshold inside a shell around the labeled skeleton flags 52 of the 802 records. One of the two radiology resident physicians read all 52 and confirmed 11, rejecting 41 (79%) as contrast, calcification and reconstruction artefact. Attenuation does not separate the two groups (both saturate); site and volume do, every confirmed implant at 2,586 mm³ or above and every rejection at 1,768 mm³ or below. Instrumentation prevalence is 1.4%, not the 6.5% the threshold alone reports.

II.I.0.b. The subtype block had to be extended. Identifiers 76–79 name spinal instrumentation (generic, cage, screw-and-rod, plate) and cannot describe what this cohort holds, so **80 arthroplasty**, **81 sacroiliac screw** and **82 osteosynthesis** were added: osteosynthe-

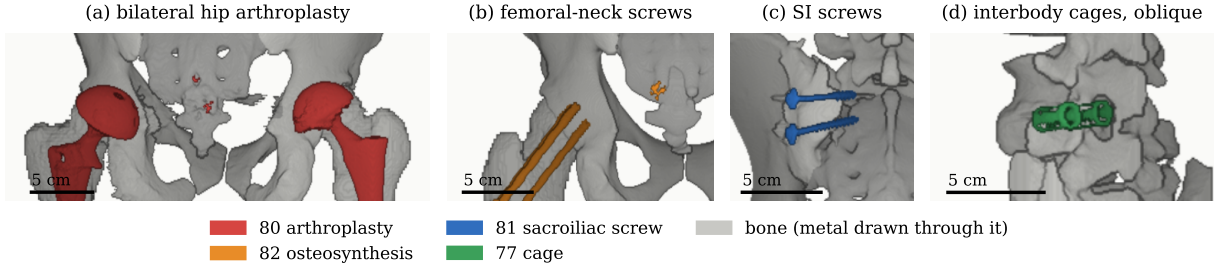


FIG. 3 Instrumentation in the release, one exemplar each, drawn through bone. (a) Hip arthroplasty, 80, $n = 8$. (b) Femoral-neck screws, 82, $n = 1$. (c) Sacroiliac screws, 81, $n = 1$. (d) Interbody cages, 77, $n = 1$. Bar, 5 cm.

sis holds parts of one bone together where arthroplasty replaces a joint, and a shape rule
calls a femoral stem a rod. The confirmed cases are 8 arthroplasties, one osteosynthesis, one
sacroiliac fixation and one pair of interbody cages (Fig. 3).

II.I.0.c. Metal outranks bone, and that is a subtraction. In every confirmed case the
implant was already inside a bone label, so naming it reclaims 1,538,852 voxels rather than
adding them. Pelvic incidence and tilt are measured from the femoral head center, and in
8 of these cases that head is an implant: any spinopelvic parameter previously computed
from them was measured on metal.

III. DATA FORMAT AND USAGE NOTES

Each record is a gzipped NIfTI image/label pair sharing an identical 4×4 affine, canon-
icalized to PIR, requiring no resampling. Masks are NIfTI rather than DICOM: the policy
prefers DICOM where applicable, and for volumetric segmentation masks NIfTI is what the
downstream tooling expects.

Splits are patient-grouped and LSTV-stratified five-fold, frozen and shipped with the
data as `splits_5fold.json`. The five validation folds partition all 802 records, so the
release provides cross-validation and *not* a held-out test set; users reporting a single held-
out number should carve one out and state which records it holds.

Stratification is on the transitional subtype rather than a binary LSTV flag: every val-
idation fold carries three sacralization and three lumbarization records. With 15 of each
across 802 that is the most stratification can guarantee, and it is why a fold-level metric on

TABLE III Data types and metadata, and the records populating each, from the released manifest.

Field	Records	Available
Identifier, image path, label path, configuration, match type	802	100.0%
Patient position, tube potential, section thickness, kernel	802	100.0%
Transitional label and class	802	100.0%
Spine and pelvic annotation origin	802	100.0%
Instrumentation and partial-annotation flags	802	100.0%
Image/label alignment check	802	100.0%
Spine series identifier; bone-fraction check	782	97.5%
Scanner manufacturer and model	768	95.8%
Sex	749	93.4%
Age	709	88.4%
Pelvic series identifier	713	88.9%
Castellvi grade (consensus of two readers)	33	4.1%

the rare classes carries an error bar far wider than its own decimal places.

Table III lists the key data types and metadata against the fraction of records for which each is populated.

The archive of record is Zenodo (<https://doi.org/10.5281/zenodo.22139642>). Code and documentation are on GitHub at the tagged release commit; any model-hub copy is a convenience mirror, not the archive.

IV. DESCRIPTIVE ANALYSIS

The cohort is a colorectal cancer screening population of 802 records. 738 are recorded female (393) or male (345); 11 carry DICOM’s *other* value, a recorded value and not a missing one, and 53 carry no sex field. Age is present for 709 (median 59, range 50–89). All 802 are counted in totals; the eleven records outside the female/male dichotomy are excluded from sex-stratified measures rather than folded into “unrecorded”. The lower age bound is the screening guideline, not a selection applied here.

Count-free measures (Fig. 4). Five rib-free vertebrae above the sacrum is typical; four

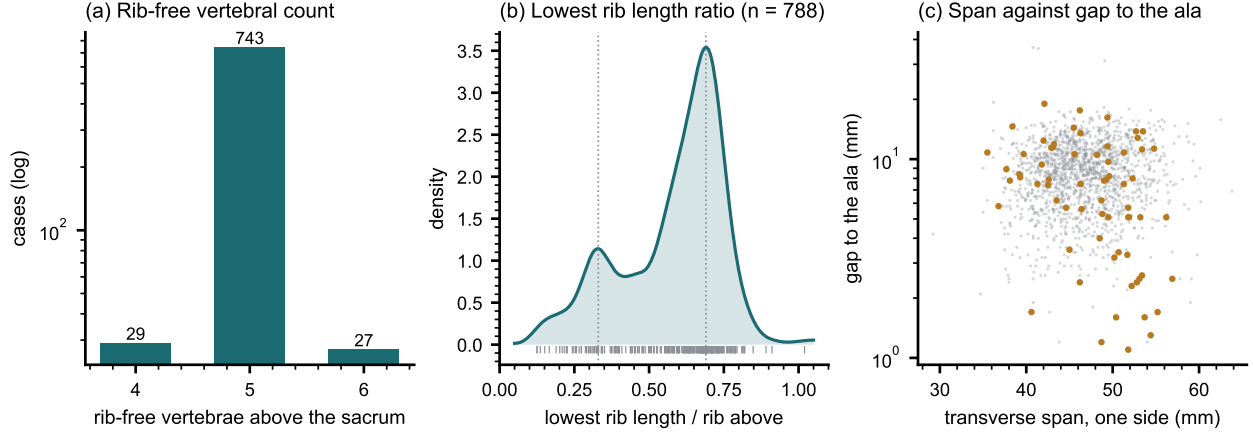


FIG. 4 Count-free measures. (a) Vertebrae between the lowest rib and the sacrum. (b) Lowest rib length over the rib above. (c) Lowest lumbar transverse span against its gap to the ala; transitional labels marked.

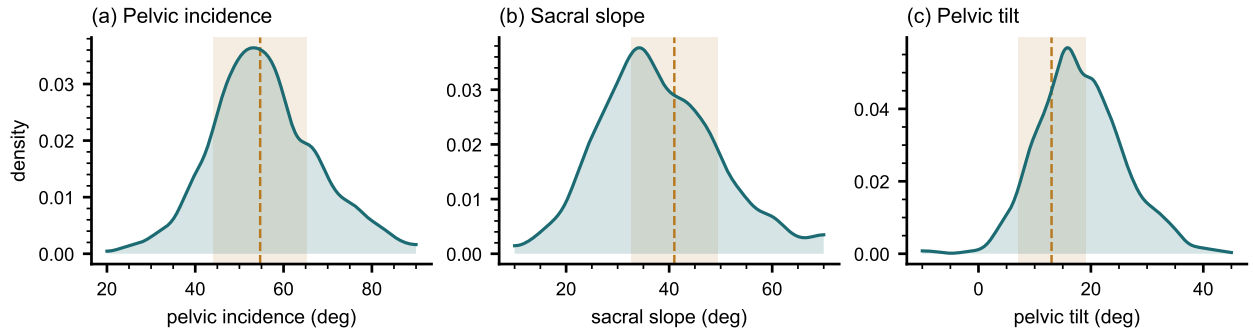


FIG. 5 Spinopelvic measures against standing reference bands (mean $\pm$ SD)²³.

and six are where transitional anatomy sits, and which of the two is present does not by itself determine what to call it. The lowest-rib length ratio is bimodal.

Level gradients and spinopelvic measures (Fig. 5). Vertebral dimensions increase caudally in step with the load each level carries. Pelvic incidence does not vary with age, while pelvic tilt increases, the descriptive signature of a pelvis retroverting as a spine ages.

Opportunistic measures. Every scan carries a vertebral bone density measurement at no additional dose.²⁵ L1 trabecular attenuation is reported at the published standard site.

V. POTENTIAL APPLICATIONS AND LIMITATIONS

The principal use is the *spinopelvic interface*. Spine and pelvis carry radiologist-sourced annotation on one coordinate frame in 802 records, making sacral morphology measurable

328 against the lumbar column rather than in isolation: the sacral endplate and its slope, the
329 alar corridors constraining S2-alar-iliac and iliac screw trajectories, and the L5–S1 geometry
330 a lumbosacral fusion must cross. Pelvic incidence is derived from the segmented sacrum and
331 femoral heads rather than from landmarks placed on a radiograph, so it is reproducible from
332 the labels themselves.

333 The transitional layer serves that use rather than competing with it: no trajectory can be
334 planned across a junction whose segments cannot be named. Level-numbering benchmarks,
335 variant studies and opportunistic screening follow from the same annotation.

336 **V.A. Three uses the released measurements support**

337 *Naming a level from its own geometry.* Wrong-level spine surgery runs at roughly one
338 in 3,110 spinal procedures, and the cause most often cited is the transitional anatomy this
339 cohort was assembled around.²⁶ The failure is structural rather than careless: the count
340 is ambiguous exactly where the variant sits. Per-level body height, canal width, endplate
341 width, transverse-process span and wedge ratio are released for T11–L5 across all 802 records,
342 and because the labels were assigned against the twelfth rib rather than by enumeration, a
343 model trained on them is not learning the convention that fails.

344 *Reference morphometry with its spread.* The values a surgeon consults for vertebral
345 body height, canal dimensions and pedicle width come from cadaveric series of a few dozen
346 specimens, plotted as single values with no indication of spread.¹⁴ They cannot tell a reader
347 whether the patient in front of them is unusual, because they never show what usual looks
348 like. The same measures are given here from 802 records with the distribution attached
349 (Fig. 6), reproducing two properties of the classical figures (body height crosses over, dorsal
350 exceeding ventral at T11–T12 and reversing by L4–L5, and pedicle width widens steeply
351 into the lower lumbar spine) and adding the width of each distribution, which is the part
352 needed to judge an individual. A larger n sharpens a distribution without changing the
353 population it came from, and this one is a screening cohort aged 50 and over.

354 *Patient-specific models.* Planning is moving toward patient-specific biomechanical models
355 rather than population norms.²⁷ Such a model needs spine, pelvis and femoral heads in
356 one frame, which is this release’s construction. In 351 patients the annotation sits on two
357 acquisitions in different positions, giving a within-patient change in alignment against which

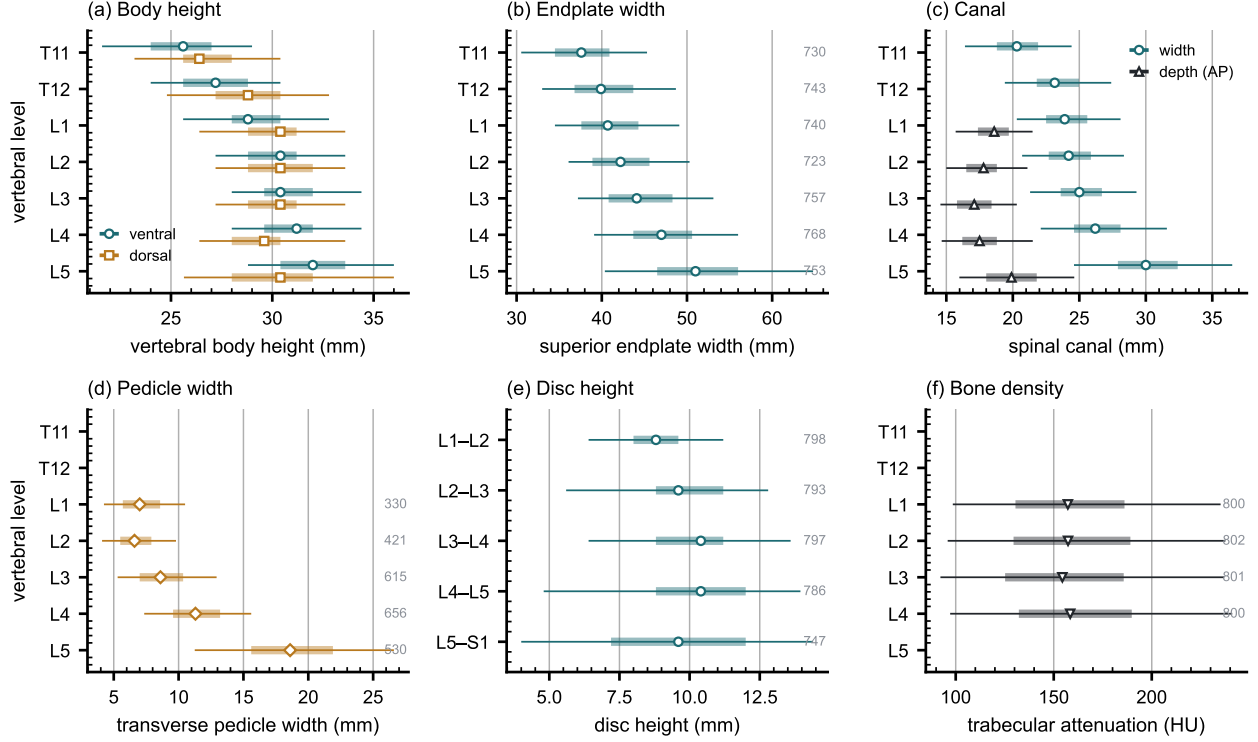


FIG. 6 Morphometry by level: median, interquartile range, 5th–95th percentile, n at right. (a) Body height. (b) Endplate width. (c) Canal. (d) Pedicle width. (e) Disc height. (f) Trabecular attenuation.

a predicted postural response can be checked.

Its limitations are stated at the level of detail a reader would need to catch them independently.

Coverage. Thoracic ground truth is field-of-view limited (Fig. 7) and does not extend to T1. Postural angles are supine; pelvic incidence is not postural and needs no such caveat. The cohort is a colorectal screening population aged 50 and over, so its distributions should not be read as representative of a surgical one.

Declared but empty, and confirmed so. The partial-annotation sentinel (255) is declared in the scheme and occurs in no record, established by counting identifiers across all 802 released volumes rather than asserted. It is retained because the protocol is part of the scheme and a later release may ship a partially traced record. The hardware identifiers were in this position until v6 and are no longer: 76–82 are populated in the eleven instrumented records described in Sec. II.I.

Strength of the labels varies by structure, and the release does not average over that.

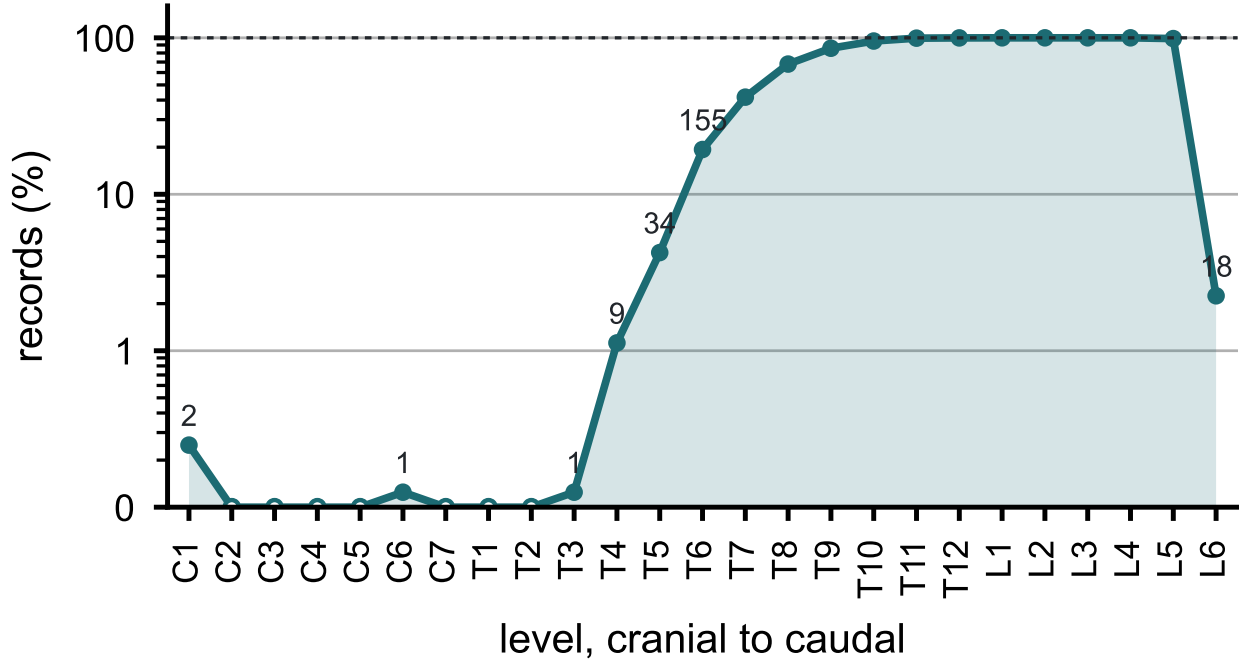


FIG. 7 Records carrying each vertebral level, cranial to caudal. Counts per identifier in Table S1.

Vertebral labels derive from radiologist-supervised source annotations, corrected where those sources were wrong. Pelvic labels on records that lacked one are pseudolabeled. The rib layer is a pseudolabel whose human review was triaged by an automated rule rather than exhaustive, so its guarantee is the absence of the failure modes that rule detects. Transitional labels derive from two independent sources and are not uniformly adjudicated, which is precisely why the measures reported here are count-free, and the Castellvi grades are a consensus of two radiology resident physicians. The sacral sub-division inherits its level from an automatic method, and in 100 records that level places S1 above half the sacrum or below fifteen percent of it, which no first sacral segment is; those records are flagged in the released quality-control table and should be filtered before any measurement depending on the sacral endplate.

Structures are not universally present. The same census shows the release is not uniformly complete. Sacrum, hips and femora are present in all 802 records. One record carries no S1, one lacks T12, and nine lack an L5 identifier, in every case because the structure lies outside the field of view or, for L5, because the lowest lumbar segment is labeled L6 or incorporated into the sacrum. Users filtering on the presence of the caudal anchor should filter explicitly rather than assume it.

389 VI. DISCUSSION

390 *Comparison with related material.* Against the collections in Table I, the contribution
391 is the crosswalk placing spine and pelvis on the same series, together with classes for the
392 anatomy an enumeration anomaly actually produces. VerSe⁶ comes closest in intent: it
393 also enriched its cohort for transitional anatomy and graded each case by Castellvi. The
394 difference is what a graded case leaves behind. VerSe names the anomaly but does not
395 segment the sacrum, so the fused or articulating structure the grade describes has no mask
396 to go with it. This release segments that structure directly, as L6 or as a separately carved
397 S1, so a user has the anatomy itself to measure rather than only a grade that describes it.
398 TotalSegmentator²⁰ invites a different comparison, since it segments far more of the body
399 than any single source used here. But its label scheme simply has no class for a sixth lumbar
400 vertebra or for a rib on a lumbar body. A six-lumbar spine cannot be recorded in it: the
401 column must be renumbered or a level dropped to fit. What this release adds, then, is not
402 more anatomy per scan (TotalSegmentator already covers more), but the specific classes an
403 enumeration anomaly needs in order to be recorded accurately.

404 *Limitations.* The cohort comes from one acquisition protocol of a colorectal-screening
405 population aged 50 and over, so how well it generalizes to a pathologic population is untested.
406 The rib layer’s human review was triaged by an automated rule rather than performed on
407 every bone, so it guarantees only the absence of the failure modes that rule detects.

408 *Future directions.* The version history shows the scheme being extended more than once:
409 femora and the S1 anchor, then per-level ribs, then lumbar ribs and surgical hardware, were
410 each folded into the existing release rather than requiring a new dataset. The same path is
411 open going forward: imaging from beyond this cohort’s single protocol, and the C2 anchor
412 this acquisition cannot supply, are the natural next additions.

413 VII. CONCLUSION

414 CTSpinoPelvic1K places spine, pelvis, per-level ribs and femora on one coordinate frame
415 in 802 records, and gives the anatomy that makes lumbar numbering ambiguous a class of
416 its own rather than recording it as something it is not. Because both counting anchors are
417 explicit, a level can be identified from a field of view that cannot support the conventional

418 count downward from C2, which is every record here.

419 DATA AVAILABILITY

420 The release described here is v8, permanently archived at [https://doi.org/10.5281/](https://doi.org/10.5281/zenodo.22293878)
421 [zenodo.22293878](https://doi.org/10.5281/zenodo.22293878); the concept DOI <https://doi.org/10.5281/zenodo.22139642> resolves
422 to the current version. The archive carries the loader, the label scheme, the quality-control
423 tables and the analysis code under an open-source licence. The repository URL identifies
424 the authors; it is on the title page and will be restored here for the camera-ready version.

425 ACKNOWLEDGEMENTS

426 Acknowledgements are given on the title page, per the journal’s double-anonymized re-
427 view policy. This work received no funding.

428 CONFLICT OF INTEREST

429 The authors have no relevant conflicts of interest to disclose.

430 ETHICS

431 This work uses publicly available, de-identified imaging and annotations derived from it.
432 The authors’ institutional review board determined that it is not human participant research
433 under 45 CFR 46 and requires no oversight.

434 REFERENCES

- 435 ¹J. Lian, N. Levine, and W. Cho, A review of lumbosacral transitional vertebrae and
436 associated vertebral numeration, *Eur. Spine J.* **27**, 995–1004 (2018), doi:10.1007/s00586-
437 018-5554-8.
- 438 ²G. P. Konin and D. M. Walz, Lumbosacral transitional vertebrae: classification, imag-
439 ing findings, and clinical relevance, *AJNR Am. J. Neuroradiol.* **31**, 1778–1786 (2010),
440 doi:10.3174/ajnr.A2036.

- ³Y. Deng et al., CTSpine1K: a large-scale dataset for spinal vertebrae segmentation in computed tomography, *Mach. Learn. Biomed. Imaging* **3**, 824–832 (2025), doi:10.59275/j.melba.2025-gf84.
- ⁴P. Liu et al., Deep learning to segment pelvic bones: large-scale CT datasets and baseline models, *Int. J. Comput. Assist. Radiol. Surg.* **16**, 749–756 (2021), doi:10.1007/s11548-021-02363-8.
- ⁵H. Möller, H. Schoen, R. Graf, et al., VERIDAH: solving enumeration anomaly aware vertebra labeling across imaging sequences, 2026, arXiv:2601.14066. doi:10.48550/arXiv.2601.14066.
- ⁶A. Sekuboyina et al., VerSe: a vertebrae labelling and segmentation benchmark for multi-detector CT images, *Med. Image Anal.* **73**, 102166 (2021), doi:10.1016/j.media.2021.102166.
- ⁷H. Liebl et al., A computed tomography vertebral segmentation dataset with anatomical variations and multi-vendor scanner data, *Sci. Data* **8**, 284 (2021), doi:10.1038/s41597-021-01060-0.
- ⁸Y. Otake, S. Schafer, J. W. Stayman, et al., Automatic localization of vertebral levels in x-ray fluoroscopy using 3D-2D registration: a tool to reduce wrong-site surgery, *Phys. Med. Biol.* **57**, 5485–5508 (2012), doi:10.1088/0031-9155/57/17/5485.
- ⁹S.-F. L. Lo, Y. Otake, V. Puvanesarajah, et al., Automatic localization of target vertebrae in spine surgery: clinical evaluation of the LevelCheck registration algorithm, *Spine (Phila. Pa. 1976)* **40**, E476–E483 (2015), doi:10.1097/BRS.0000000000000814.
- ¹⁰T. De Silva, S.-F. L. Lo, N. Aygun, et al., Utility of the LevelCheck algorithm for decision support in vertebral localization, *Spine (Phila. Pa. 1976)* **41**, E1249–E1256 (2016), doi:10.1097/BRS.0000000000001589.
- ¹¹M. R. Zindrick et al., Analysis of the morphometric characteristics of the thoracic and lumbar pedicles, *Spine (Phila. Pa. 1976)* **12**, 160–166 (1987), doi:10.1097/00007632-198703000-00012.
- ¹²M. M. Panjabi et al., Thoracic human vertebrae: quantitative three-dimensional anatomy, *Spine (Phila. Pa. 1976)* **16**, 888–901 (1991), doi:10.1097/00007632-199108000-00006.
- ¹³M. M. Panjabi et al., Human lumbar vertebrae: quantitative three-dimensional anatomy, *Spine (Phila. Pa. 1976)* **17**, 299–306 (1992), doi:10.1097/00007632-199203000-00010.
- ¹⁴E. C. Benzal, *Biomechanics of Spine Stabilization*, Thieme, New York, 3rd edition, 2015.

¹⁵R. J. Hughes and A. Saifuddin, Numbering of lumbosacral transitional vertebrae on MRI: role of the iliolumbar ligaments, *AJR Am. J. Roentgenol.* **187**, W59–W65 (2006), doi:10.2214/AJR.05.0415.

¹⁶L. Jin et al., RibSeg v2: a large-scale benchmark for rib labeling and anatomical centerline extraction, *IEEE Trans. Med. Imaging* **43**, 570–581 (2024), doi:10.1109/TMI.2023.3313627.

¹⁷K. Smith et al., Data from CT COLONOGRAPHY, The Cancer Imaging Archive, 2015, doi:10.7937/K9/TCIA.2015.NWTESAY1.

¹⁸K. Clark et al., The Cancer Imaging Archive (TCIA): maintaining and operating a public information repository, *J. Digit. Imaging* **26**, 1045–1057 (2013), doi:10.1007/s10278-013-9622-7.

¹⁹H. Möller et al., Automated thoracolumbar stump rib detection and analysis in a large CT cohort, 2025, arXiv:2505.05004. doi:10.48550/arXiv.2505.05004.

²⁰J. Wasserthal et al., TotalSegmentator: robust segmentation of 104 anatomic structures in CT images, *Radiol. Artif. Intell.* **5**, e230024 (2023), doi:10.1148/ryai.230024.

²¹[Reference to the authors’ own work, removed for double-anonymized review.]

²²F. Isensee, P. F. Jaeger, S. A. A. Kohl, J. Petersen, and K. H. Maier-Hein, nnU-Net: a self-configuring method for deep learning-based biomedical image segmentation, *Nat. Methods* **18**, 203–211 (2021), doi:10.1038/s41592-020-01008-z.

²³R. Vialle, N. Levassor, L. Rillardon, A. Templier, W. Skalli, and P. Guigui, Radiographic analysis of the sagittal alignment and balance of the spine in asymptomatic subjects, *J. Bone Joint Surg. Am.* **87**, 260–267 (2005), doi:10.2106/JBJS.D.02043.

²⁴A. E. Castellvi, L. A. Goldstein, and D. P. K. Chan, Lumbosacral transitional vertebrae and their relationship with lumbar extradural defects, *Spine (Phila. Pa. 1976)* **9**, 493–495 (1984), doi:10.1097/00007632-198407000-00014.

²⁵P. J. Pickhardt, B. D. Pooler, T. Lauder, A. M. del Rio, R. J. Bruce, and N. Binkley, Opportunistic screening for osteoporosis using abdominal computed tomography scans obtained for other indications, *Ann. Intern. Med.* **158**, 588–595 (2013), doi:10.7326/0003-4819-158-8-201304160-00003.

²⁶N. E. Epstein, A perspective on wrong level, wrong side, and wrong site spine surgery, *Surg. Neurol. Int.* **12**, 286 (2021), doi:10.25259/SNI.402_2021.

⁵⁰⁴ ²⁷E. Beaulieu et al., A review of finite element analysis in spine surgery decision-making, J.
⁵⁰⁵ Clin. Med. **15**, 2584 (2026), doi:10.3390/jcm15072584.

Supplementary Material

CTSpinoPelvic1K: spine, pelvis, ribs and femora in one coordinate frame, annotated for lumbosacral transitional anatomy

Table S1: Every identifier populated in the release and the number of the 802 records carrying it, counted from the released label volumes. Identifiers 58–73 are unassigned and occur nowhere.

id	class	records	%	id	class	records	%
1	C1	2	0.2	37	rib_left_4	65	8.1
6	C6	1	0.1	38	rib_left_5	316	39.4
10	T3	1	0.1	39	rib_left_6	635	79.2
11	T4	9	1.1	40	rib_left_7	780	97.3
12	T5	34	4.2	41	rib_left_8	800	99.8
13	T6	155	19.3	42	rib_left_9	802	100.0
14	T7	335	41.8	43	rib_left_10	802	100.0
15	T8	545	68.0	44	rib_left_11	802	100.0
16	T9	688	85.8	45	rib_left_12	787	98.1
17	T10	765	95.4	48	rib_right_3	5	0.6
18	T11	798	99.5	49	rib_right_4	62	7.7
19	T12	801	99.9	50	rib_right_5	310	38.7
20	L1	802	100.0	51	rib_right_6	624	77.8
21	L2	802	100.0	52	rib_right_7	773	96.4
22	L3	802	100.0	53	rib_right_8	800	99.8
23	L4	802	100.0	54	rib_right_9	801	99.9
24	L5	793	98.9	55	rib_right_10	801	99.9
25	L6	18	2.2	56	rib_right_11	802	100.0
26	sacrum	802	100.0	57	rib_right_12	788	98.3
29	S1	801	99.9	74	rib_left_lumbar	13	1.6
30	left_hip	802	100.0	75	rib_right_lumbar	16	2.0
31	right_hip	802	100.0	77	hardware_cage	1	0.1
32	femur_left	802	100.0	80	hardware_arthroplasty	8	1.0
33	femur_right	802	100.0	81	hardware_si_screw	1	0.1
36	rib_left_3	5	0.6	82	hardware_osteosynthesis	1	0.1